FA 8

1 (6 pts) An urn contains 10 red, 8 black, and 7 green balls. One of the color is chosen at random (meaning the chosen color is equally likely to be any of the 3 colors), and then 4 balls are randomly chosen from the urn.

Let X be the number of these balls that are of the chosen color.

Given: 10 red, 8 black, & 7 green balls, Total of 25 balls.

a) (2 pts) Find $P(X=0)$

Condition on the chosen color.

IF Red is chosen: (Non Red balls: $8+7=15$)

$$P(X=0|R) = \frac{\binom{15}{4}}{\binom{25}{4}} = \frac{1365}{12650}$$

IF Black is chosen: (Non-Black: $10+7=17$)

$$P(X=0|B) = \frac{\binom{17}{4}}{\binom{25}{4}} = \frac{2380}{12650}$$

IF Green is chosen: (Non-Green: $10+8=18$)

$$P(X=0|G) = \frac{\binom{18}{4}}{\binom{25}{4}} = \frac{3060}{12650}$$

Since each color is chosen w/ probability $\frac{1}{3}$.

$$P(X=0) = \frac{1}{3}\left(\frac{\binom{15}{4}+\binom{17}{4}+\binom{18}{4}}{\binom{25}{4}}\right)$$

$$P(X=0) = \frac{1}{3}\left[\frac{1365+2380+3060}{12650}\right] = \frac{6805}{37950} = \frac{1361}{7590} \approx 0.1793$$

b) Find $P(X_i=1)$

$X_i = 1$ if the i-th selected ball has the chosen color.

$$P(X_i=1) = \frac{1}{3}\left(\frac{10}{25}+\frac{8}{25}+\frac{7}{25}\right)$$

$$P(X_i=1) = \frac{1}{3}\left(\frac{25}{25}\right) = \frac{1}{3}$$

For every $i = 1,2,3,4$

$$\boxed{P(X_i=1) = \frac{1}{3}}$$

c) Find $E[X]$

$$X = X_1+X_2+X_3+X_4$$

$$E[X] = \sum_{i=1}^{4} E[X_i]$$

$$E[X_i] = P(X_i=1) = \frac{1}{3}$$

$$\boxed{E[X] = 4\left(\frac{1}{3}\right) = \frac{4}{3}}$$

2 (8 pts) The random variable X is said to have the Yule-Simon distribution if

$$P(X=n) = \frac{4}{n(n+1)(n+2)}, \quad n \geq 1.$$

a) (3 pts) Show that the preceding is actually a probability mass function. That is, show that

$$\sum_{n=1}^{\infty} P(X=n) = 1$$

$$\frac{1}{n(n+1)(n+2)} = \frac{1}{2}\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right]$$

$$P(X=n) = 2\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right]$$

Summing from $n=1$ to N

$$\sum_{n=1}^{N} P(X=n) = 2\sum_{n=1}^{N}\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right]$$

Telescoping:

$$= 2\left(\frac{1}{1\cdot 2} - \frac{1}{(N+1)(N+2)}\right)$$

$N \to \infty$

$$\sum_{n=1}^{\infty} P(X=n) = 2\left(\frac{1}{2}\right) = 1$$

$$\boxed{\sum_{n=1}^{\infty} P(X=n) = 1}$$

b) Show that $E[X] = 2$

$$E[X] = \sum_{n=1}^{\infty} n\,\frac{4}{n(n+1)(n+2)} = 4\sum_{n=1}^{\infty}\frac{1}{(n+1)(n+2)}$$

$$\frac{1}{(n+1)(n+2)} = \frac{1}{n+1} - \frac{1}{n+2}$$

Telescoping: $E[X] = 4\left(\frac{1}{2} - \lim_{N\to\infty}\frac{1}{N+2}\right) = 4\left(\frac{1}{2}\right)$

$$\boxed{E[X] = 2}$$

c.) Show the $E[X^2] = \infty$

$$E[X^2] = \sum_{n=1}^{\infty} n^2\,\frac{4}{n(n+1)(n+2)} = 4\sum_{n=1}^{\infty}\frac{n}{(n+1)(n+2)}$$

Partial fraction:

$$\frac{n}{(n+1)(n+2)} = -\frac{1}{n+1} + \frac{2}{n+2}$$

$$E[X^2] = 4\sum_{n=1}^{\infty}\left(-\frac{1}{n+1}+\frac{2}{n+2}\right) = \frac{1}{n} = \boxed{\infty}$$

3. (4 pts) The probability density function of X is given by

$$f(x) = \begin{cases} a+bx^2, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

If $E[X] = \frac{3}{5}$, find the values of a and b.

First Condition: Total probability = 1

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

Since $f(x)=0$ outside $[0,1]$,

$$\int_0^1 (a+bx^2)\,dx = 1$$

$$a\int_0^1 dx + b\int_0^1 x^2\,dx = 1$$

$$\left[a\int_0^1 dx = 1\right] + \left[b\int_0^1 x^2\,dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3}\right]$$

$$3\left[a + \frac{b}{3} = 1\right]3$$

$$3a + b = 3$$

Second Condition: Expected Value

For continuous random variable,

$$E[X] = \int_{-\infty}^{\infty} x f(x)\,dx$$

$f(x) = a+bx^2$:

$$\int_0^1 x(a+bx^2)\,dx = \frac{3}{5}$$

$$\int_0^1 xa + bx^3\,dx = \frac{3}{5}$$

$$\left[a\int_0^1 x\,dx + b\int_0^1 x^3\,dx\right] = \frac{3}{5}$$

$$a\int_0^1 x\,dx = a\left[\frac{x^2}{2}\right]_0^1 = a\,\frac{1}{2} = \frac{a}{2}$$

$$b\int_0^1 x^3\,dx = b\left[\frac{x^4}{4}\right]_0^1 = b\,\frac{1}{4} = \frac{b}{4}$$

$$20\left[\frac{a}{2}+\frac{b}{4} = \frac{3}{5}\right]20$$

$$10a + 5b = 12$$

$$\begin{cases} 3a+b=3 \\ 10a+5b=12 \end{cases}$$

$b = 3-3a$

$10a + 5(3-3a) = 12$

$10a + 15 - 15a = 12$

$-5a + 15 = 12$

$-5a = -3$

$$\boxed{a = \frac{3}{5}}$$

$b = 3-3a$

$= 3-3\left(\frac{3}{5}\right)$

$= 3 - \frac{9}{5}$

$= \frac{15}{5} - \frac{9}{5}$

$$\boxed{b = \frac{6}{5}}$$

4. (4pts) Let X be a random variable having expected value μ & variance σ^2. Find the expected value & variance of

$$Y = \frac{X-\mu}{\sigma}$$

Expected Value

$$E[Y] = E\left[\frac{X-\mu}{\sigma}\right] = \frac{1}{\sigma}(E[X]-\mu) = 0$$

$$E[Y] = 0$$

Variance:

$$\mathrm{Var}(Y) = \mathrm{Var}\left(\frac{X-\mu}{\sigma}\right) = \frac{1}{\sigma^2}\mathrm{Var}(X)$$

$$\mathrm{Var}(X) = \sigma^2$$

$$\boxed{\mathrm{Var}(Y) = 1}$$

5) (8 pts) Let X & Y be Independent Poisson random variables with parameters $\lambda_1 = 2$ and $\lambda_2 = 3$, respectively. Let $W = X+Y$

a) (3pts) Using the property of sums of independent random variables, derive the moment generating function (MGF) $M_W(t)$ of W.

MGF of a Poisson(λ)

$$M(t) = e^{\lambda(e^t-1)}$$

$$M_X(t) = e^{2(e^t-1)}$$

$$M_Y(t) = e^{3(e^t-1)}$$

$$M_W(t) = M_X(t)\,M_Y(t)$$

$$M_W(t) = e^{2(e^t-1)}e^{3(e^t-1)} = e^{5(e^t-1)}$$

$$\boxed{M_W(t) = e^{5(e^1-1)}}$$

b. (2pts) Based on the uniqueness property of MGFs, identify the probability distribution and parameter(s) of W.

The MGF $e^{5(e^t-1)}$ is the MGF of a Poisson random variable w/ parameter 5.

Therefore $\boxed{W \sim \text{Poisson}(5)}$.

c) (3pts) Determined the expected value $E[W]$ and the second moment $E[W^2]$ by successively differentiating $M_W(t)$ and evaluating it at $t=0$.

$$M_W(t) = e^{5(e^t-1)}$$

First Derivative

$$M'_W(t) = e^{5(e^t-1)}(5e^t)$$

At $t=0$

$$M'_W(0) = e^{5(e^0-1)}(5e^0) = e^{5(1-1)}(5(1)) = e^0(5) = 5$$

$$E[W] = 5$$

Second Derivative

$$M''_W(t) = (5e^t)e^{5(e^t-1)} + (5e^t)(5e^t)e^{5(e^t-1)}$$

$$M''_W(t) = e^{5(e^t-1)}\left(5e^t + 25e^{2t}\right)$$

Evaluate at $t=0$

$$M''_W(0) = e^{5(e^0-1)}\left(5e^0 + 25e^{2(0)}\right)$$

$$= e^{5(0)}(5+25) = e^0(30) = 30$$

$$E[W^2] = M''_W(0) \quad \boxed{E[W^2] = 30}$$

Kilonotes